Azure End-to-End ETL Capstone Project Report

1. Project Overview

This project demonstrates the design and implementation of a cloud-based ETL (Extract, Transform, Load) pipeline using Microsoft Azure. The solution involves ingesting raw data, transforming it into a clean and usable format, and visualizing insights using Power BI.

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2. Project Objective

The objective of this project is to:

- Build an end-to-end ETL pipeline
- Store raw and processed data in Azure Blob Storage
- Perform data transformation using Azure Data Factory
- Visualize insights using Power BI
- Implement cost-effective cloud solutions

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3. Architecture

Data Source → Azure Blob Storage (Raw) → Azure Data Factory → Transformation → Azure Blob Storage (Curated) → Power BI

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4. Dataset Information

- **Source:** Online Retail dataset
- **Format:** CSV
- **Name:** data.csv
- **Size:** < 100 MB
- **Description:** Real-world e-commerce transactions

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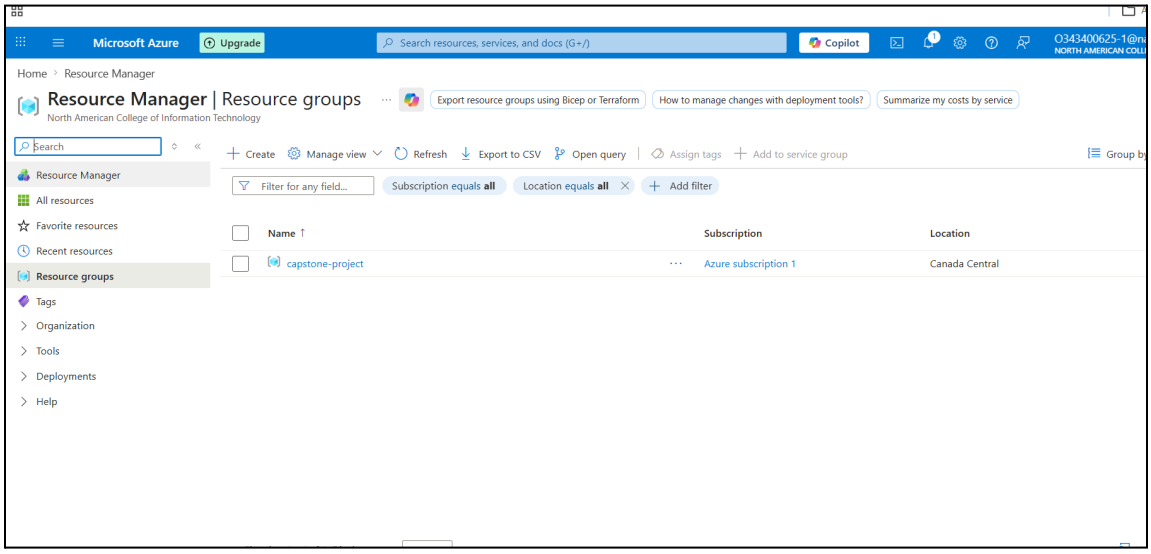
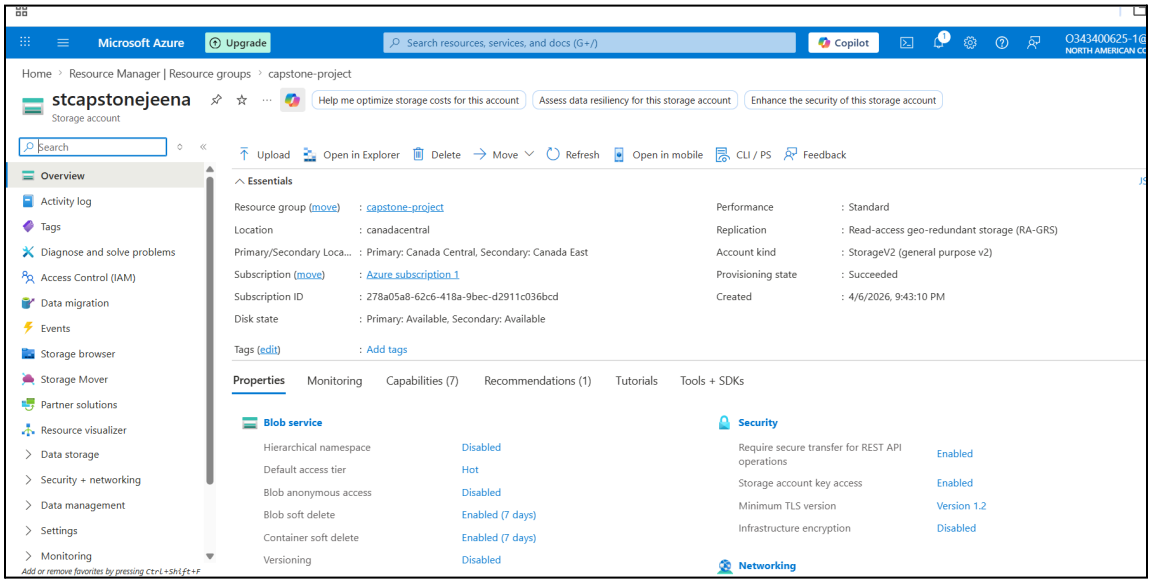
5. Azure Setup

5.1 Resource Group

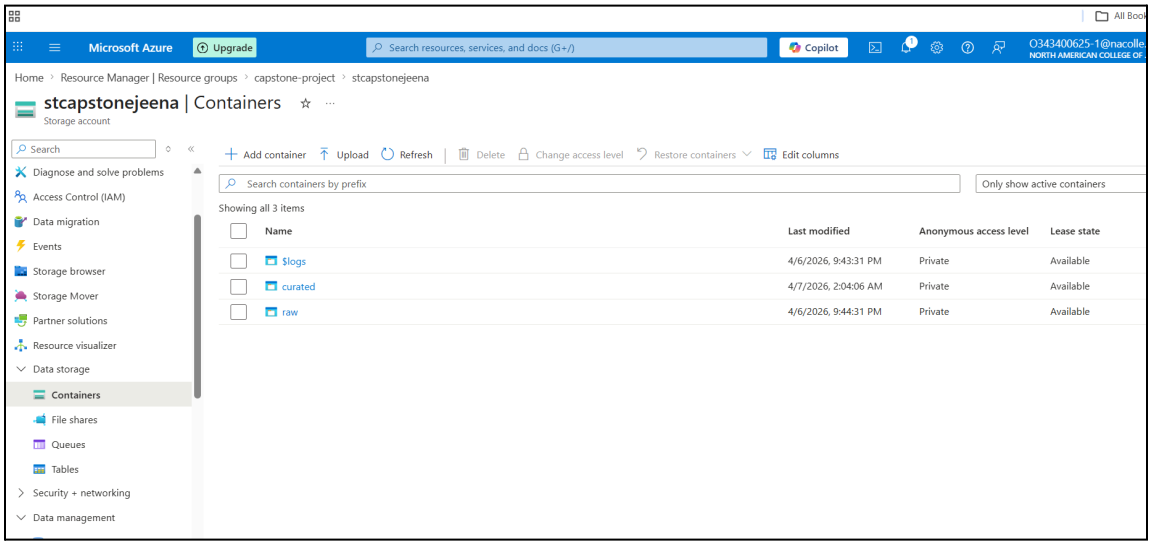


5.2 Storage Account



5.3 Blob Containers

- **Raw Container:** Stores original dataset
- **Curated Container:** Stores transformed dataset



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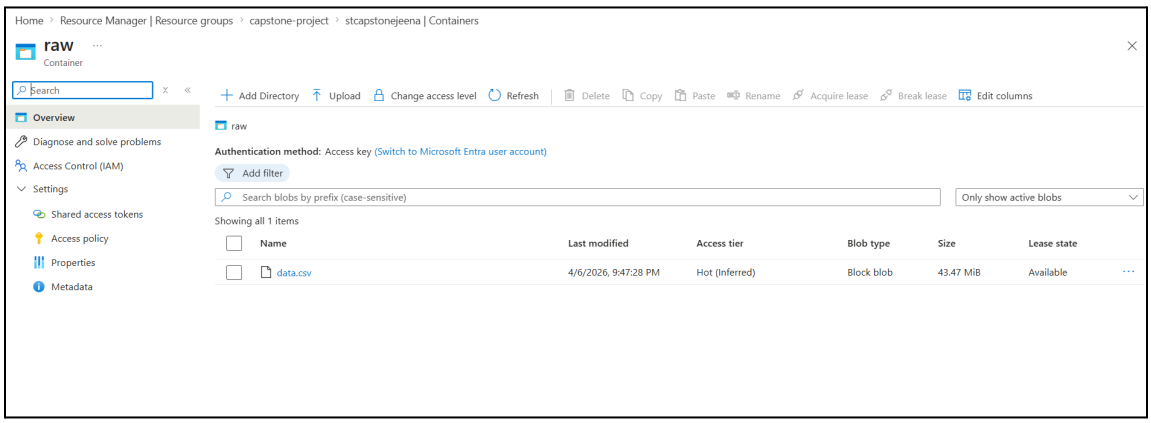
6. Data Ingestion

- Uploaded dataset to the **raw** container without modification
- Verified file upload and path



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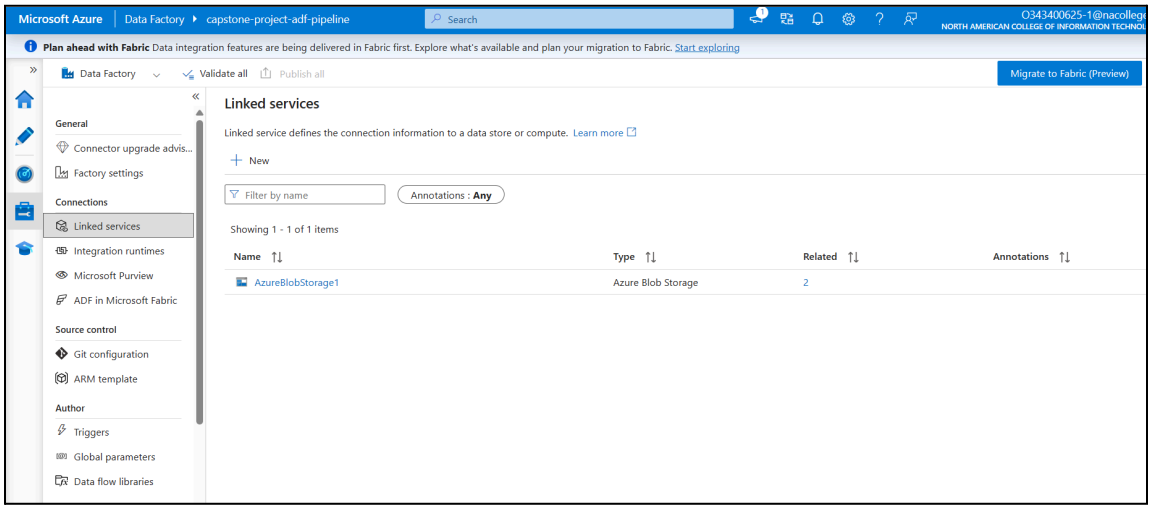
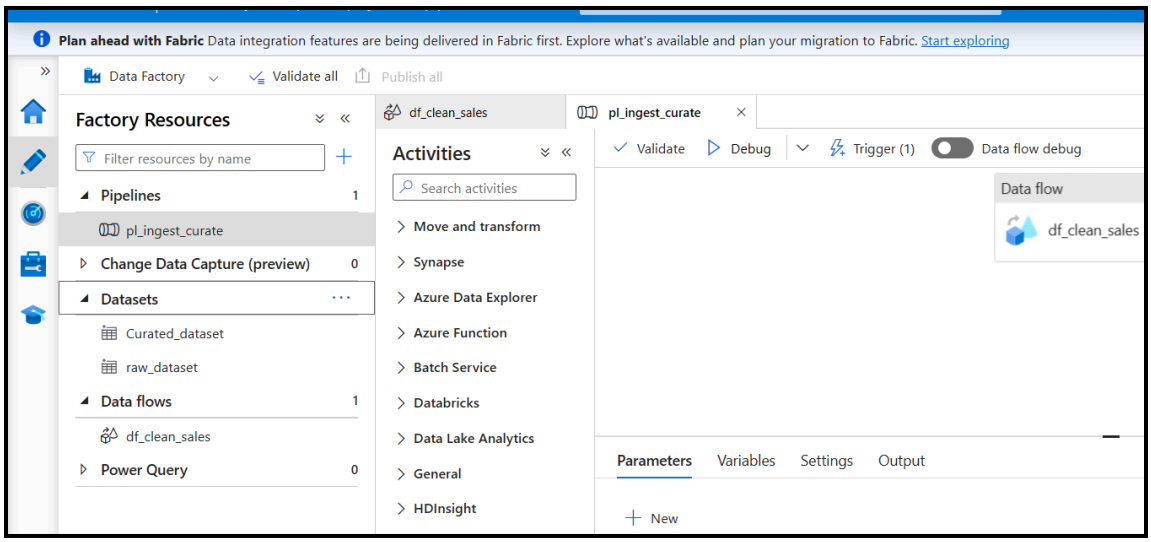
7. Azure Data Factory Setup

- Created Azure Data Factory instance
- Configured Linked Service to Blob Storage
- Created datasets for raw and curated data





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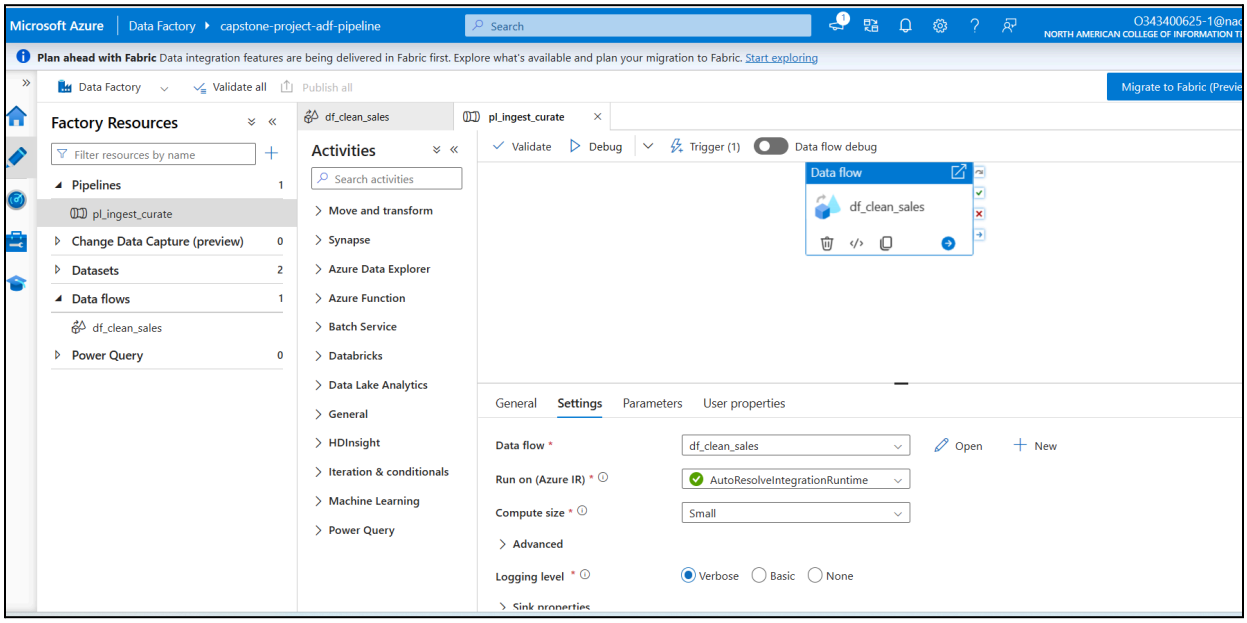
8. Pipeline Implementation

- Pipeline Name: pl_ingest_curate
- Activity Used: Copy Data / Mapping Data Flow
- Source: Raw dataset
- Sink: Curated dataset



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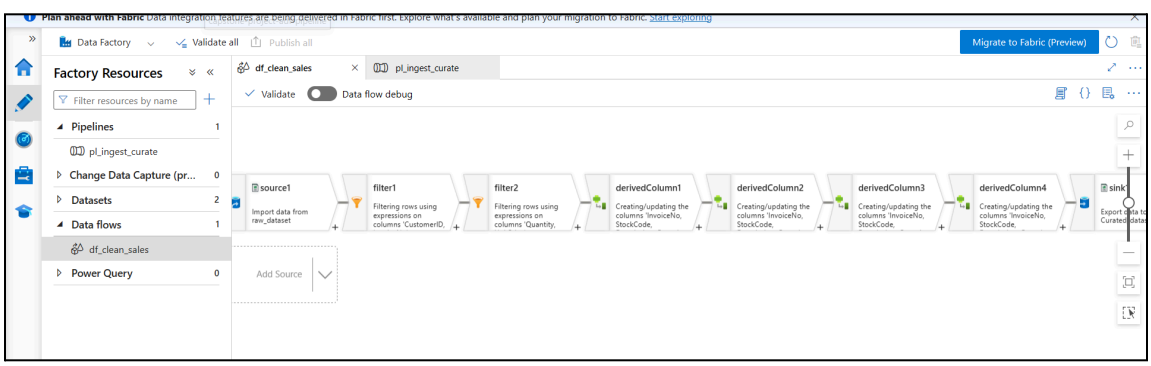
9. Data Transformation

The following transformations were applied:

- Removed null values
- Standardized data types
- Created derived column: TotalAmount = Quantity × Price
- Filtered invalid records



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10. Validation

- Pipeline execution status: Succeeded
- Verified output in curated container
- Checked row counts and column accuracy

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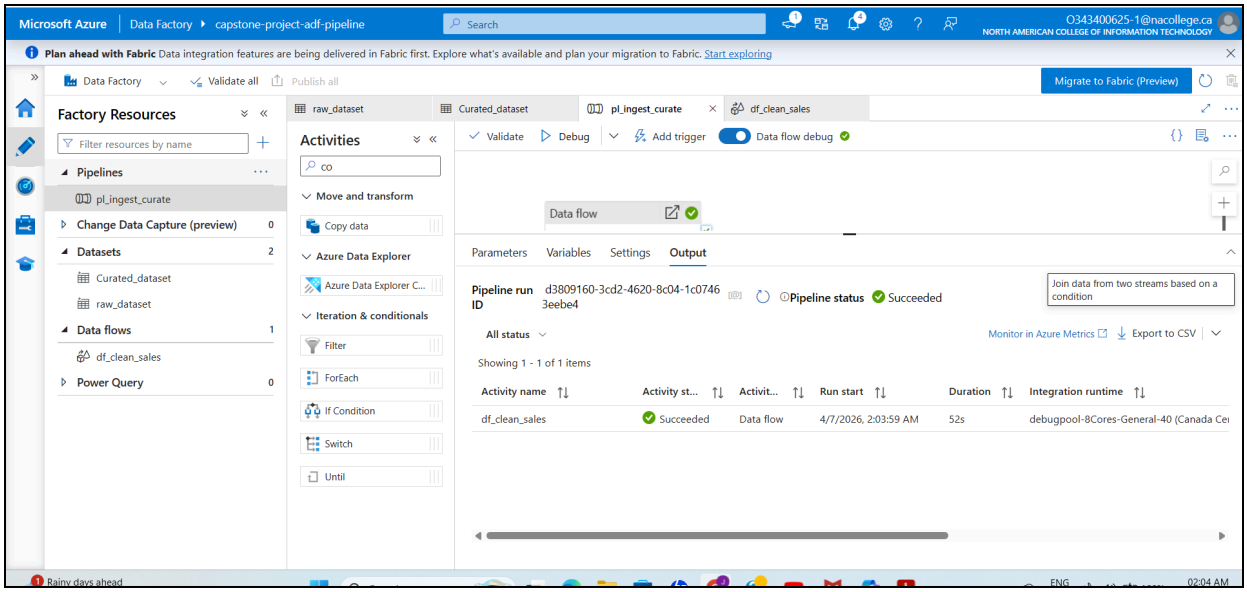
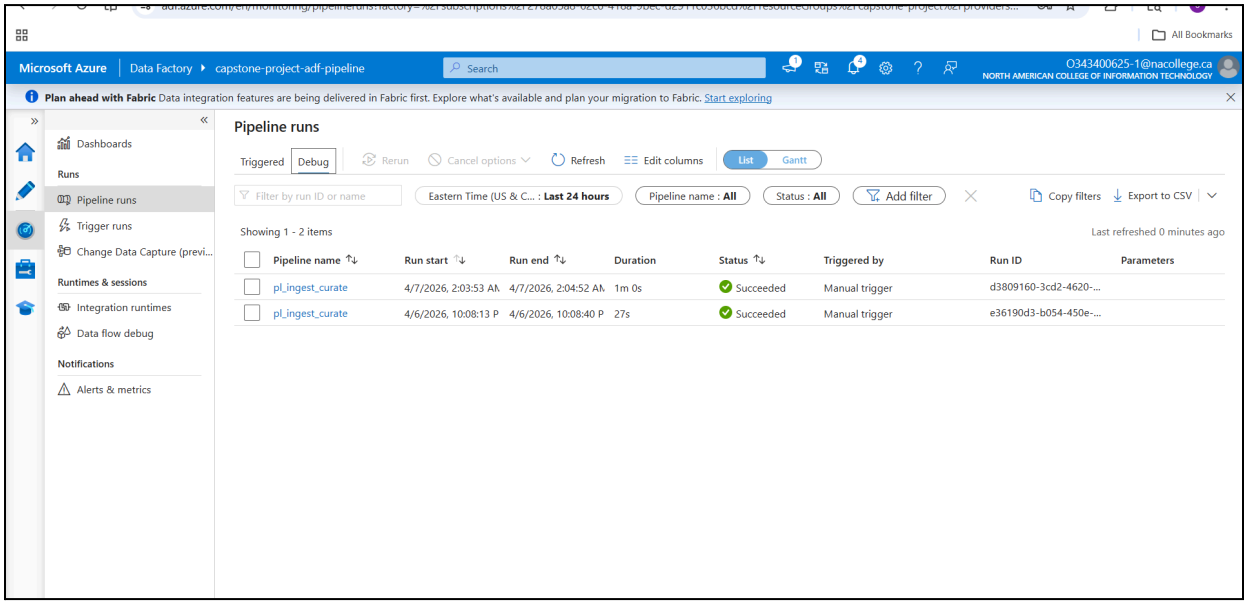
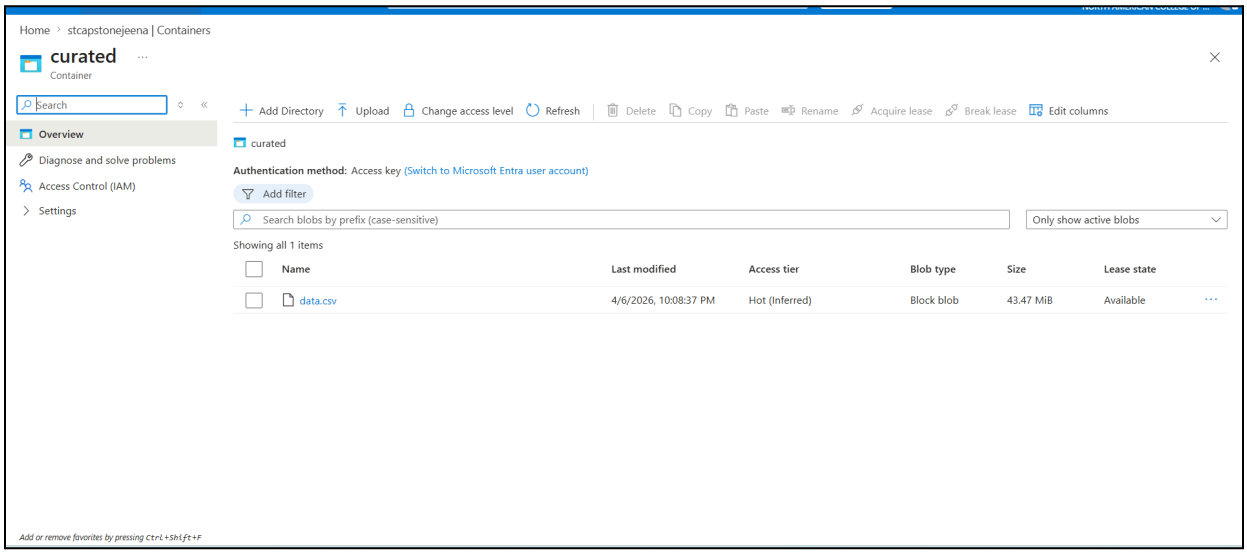
11. Monitoring

- Used ADF Monitor to track pipeline runs
- Verified execution time and status



Added Trigger





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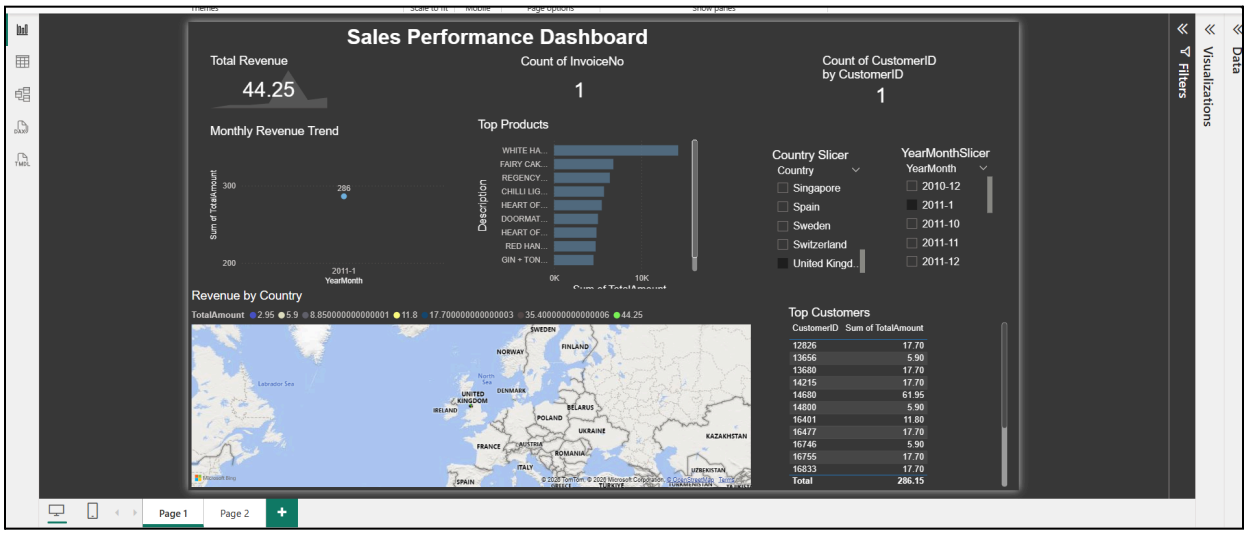
12. Power BI Dashboard

The dashboard includes:

- KPI Card: Total Revenue
- Bar Chart: Top Products by Revenue
- Line Chart: Monthly Sales Trend
- Table: Top Customers

- Slicer: country/yearMonth



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13. Insights

- Top 10 products contribute to a major portion of total revenue, indicating that sales are highly concentrated on a few high-performing items
- Sales show a clear upward trend during certain months, with peak revenue observed in 2011-11 This suggests seasonal demand patterns
- A significant portion of revenue is generated from United Kingdom, indicating strong market presence in that region
- A small group of customers contributes disproportionately to total revenue, highlighting the importance of key customers
- The dataset shows multiple transactions per customer, indicating repeat purchasing behavior
- Revenue is driven by both high-volume products and higher-priced items, suggesting a mix of bulk and premium sales.

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14. Cost Optimization

- Used small dataset
- Limited pipeline runs
- Avoided expensive Azure services
- Deleted unused resources after completion

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15. Challenges Faced

- Data type mismatches
- Handling null values
- Power BI visualization issues

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16. Solutions

- Fixed schema mapping in ADF
- Applied transformations for null handling
- Ensured correct dataset connection in Power BI

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17. Folder Structure

```
/azure-etl-capstone/
  ├── data/
  │   ├── raw/
  │   └── curated/
  ├── pipelines/
  ├── scripts/
  ├── dashboards/
  └── README.md
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18. Conclusion

This project successfully demonstrates the implementation of an end-to-end ETL pipeline using Azure services. It highlights data ingestion, transformation, orchestration, and visualization while maintaining cost efficiency.

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19. References

- Azure Documentation
- Dataset Source
- Power BI Documentation